Comparative Urban Systems

ABEE4086 Urban Systems and Techniques

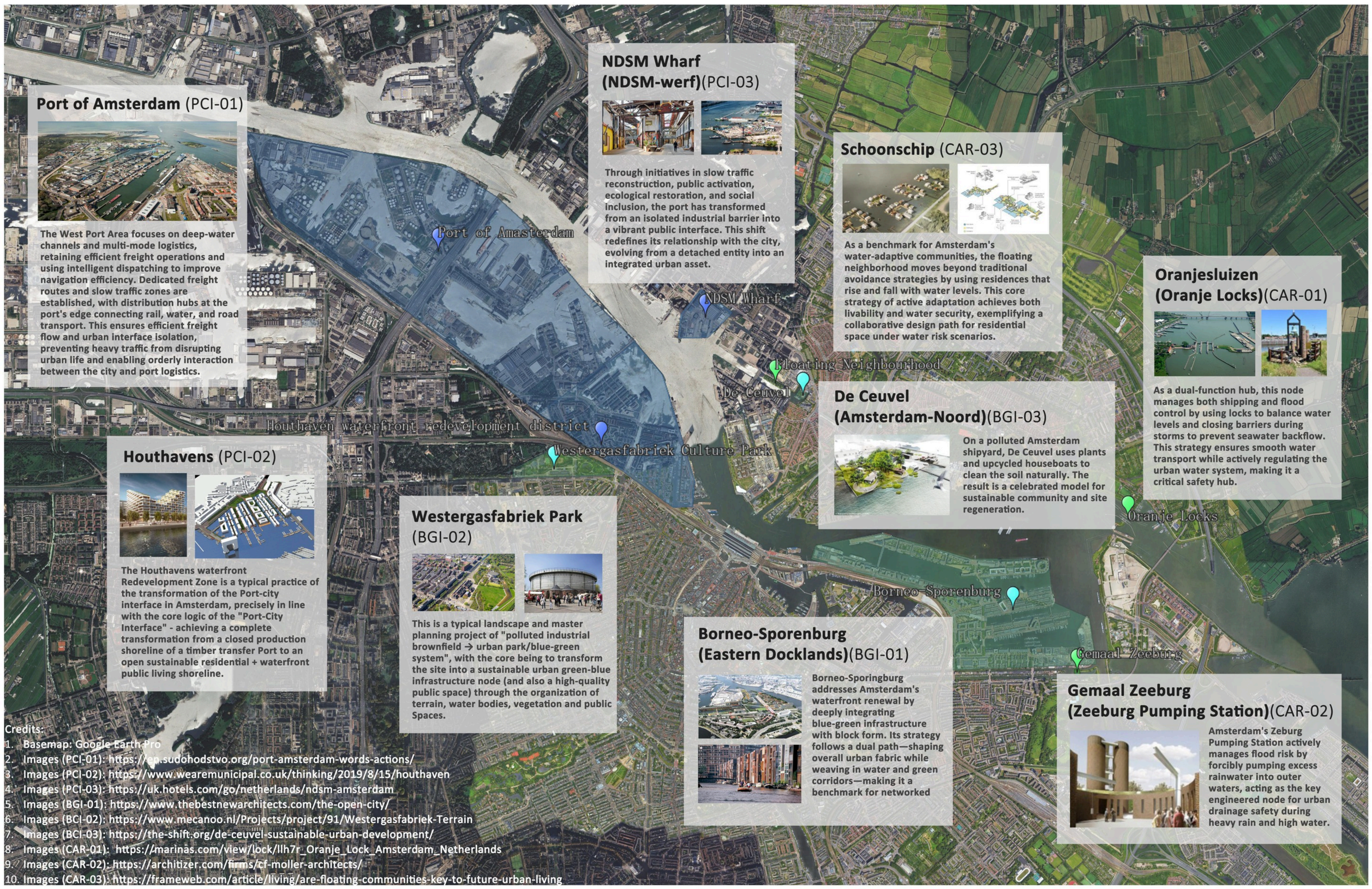
Xinyue Liu | Independent work

Dr. Alexander Zipprich [Module Convenor]

Amasterdam

Background | Introduction of Amsterdam

Amsterdam Wateredge Features



Port of Amsterdam (PCI-01)

The West Port Area focuses on deep-water channels and multi-mode logistics, retaining efficient freight operations and using intelligent dispatching to improve navigation efficiency. Dedicated freight routes and slow traffic zones are established, with distribution hubs at the port's edge connecting rail, water, and road transport. This ensures efficient freight flow and urban interface isolation, preventing heavy traffic from disrupting urban life and enabling orderly interaction between the city and port logistics.

NDSM Wharf (NDSM-werf)(PCI-03)

Through initiatives in slow traffic reconstruction, public activation, ecological restoration, and social inclusion, the port has transformed from an isolated industrial barrier into a vibrant public interface. This shift redefines its relationship with the city, evolving from a detached entity into an integrated urban asset.

Schoonschip (CAR-03)

As a benchmark for Amsterdam's water-adaptive communities, the floating neighborhood moves beyond traditional avoidance strategies by using residences that rise and fall with water levels. This core strategy of active adaptation achieves both livability and water security, exemplifying a collaborative design path for residential space under water risk scenarios.

Oranjesluizen (Oranje Locks)(CAR-01)

As a dual-function hub, this node manages both shipping and flood control by using locks to balance water levels and closing barriers during storms to prevent seawater backflow. This strategy ensures smooth water transport while actively regulating the urban water system, making it a critical safety hub.

De Ceuvel (Amsterdam-Noord)(BGI-03)

On a polluted Amsterdam shipyard, De Ceuvel uses plants and upcycled houseboats to clean the soil naturally. The result is a celebrated model for sustainable community and site regeneration.

Houthavens (PCI-02)

The Houthavens waterfront Redevelopment Zone is a typical practice of the transformation of the Port-city interface in Amsterdam, precisely in line with the core logic of the "Port-City Interface" - achieving a complete transformation from a closed production shoreline of a timber transfer Port to an open sustainable residential + waterfront public living shoreline.

Westergasfabriek Park (BGI-02)

This is a typical landscape and master planning project of "polluted industrial brownfield → urban park/blue-green system", with the core being to transform the site into a sustainable urban green-blue infrastructure node (and also a high-quality public space) through the organization of terrain, water bodies, vegetation and public Spaces.

Borneo-Sporenburg (Eastern Docklands)(BGI-01)

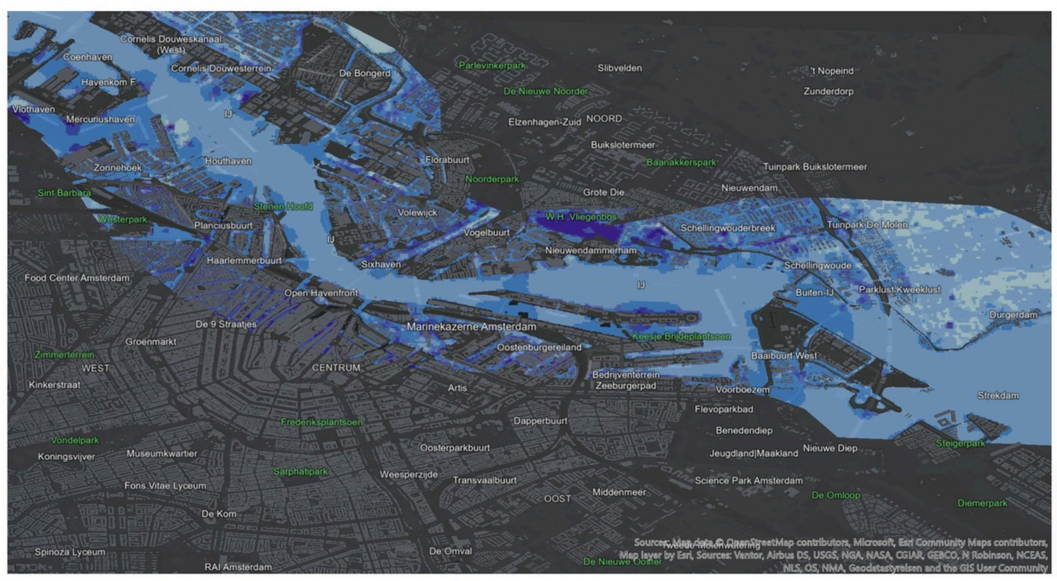
Borneo-Sporingburg addresses Amsterdam's waterfront renewal by deeply integrating blue-green infrastructure with block form. Its strategy follows a dual path—shaping overall urban fabric while weaving in water and green corridors—making it a benchmark for networked

Gemaal Zeeburg (Zeeburg Pumping Station)(CAR-02)

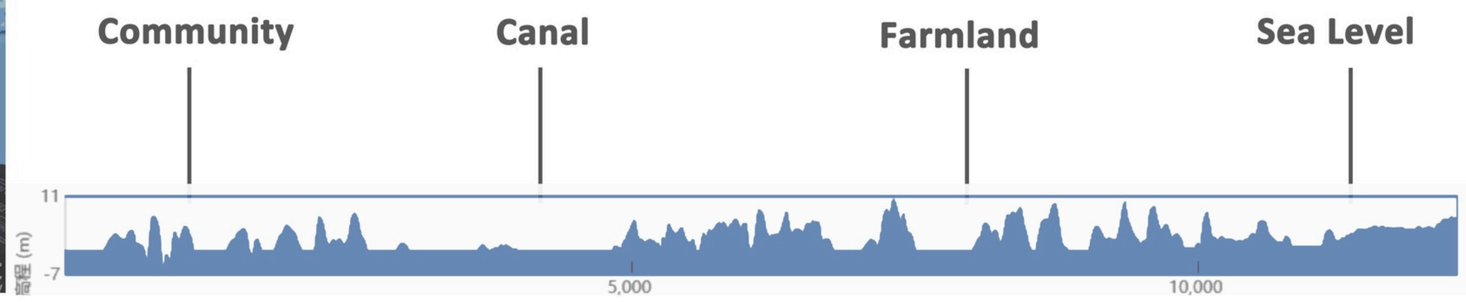
Amsterdam's Zeburg Pumping Station actively manages flood risk by forcibly pumping excess rainwater into outer waters, acting as the key engineered node for urban drainage safety during heavy rain and high water.

Credits:
1. Basemap: Google Earth Pro
2. Images (PCI-01): https://en.sudohodstvo.org/port-amsterdam-words-actions/
3. Images (PCI-02): https://www.wearemunicipal.co.uk/thinking/2019/8/15/houthaven
4. Images (PCI-03): https://uk.hotels.com/go/netherlands/ndsm-amsterdam
5. Images (BGI-01): https://www.thebestnewarchitects.com/the-open-city/
6. Images (BCI-02): https://www.mecanoo.nl/Projects/project/91/Westergasfabriek-Terrain
7. Images (BCI-03): https://the-shift.org/de-ceuvel-sustainable-urban-development/
8. Images (CAR-01): https://marinas.com/view/lock/llh7r_Oranje_Lock_Amsterdam_Netherlands
9. Images (CAR-02): https://architizer.com/firms/cf-moller-architects/
10. Images (CAR-03): https://frameweb.com/article/living/are-floating-communities-key-to-future-urban-living

Why is Amsterdam ?



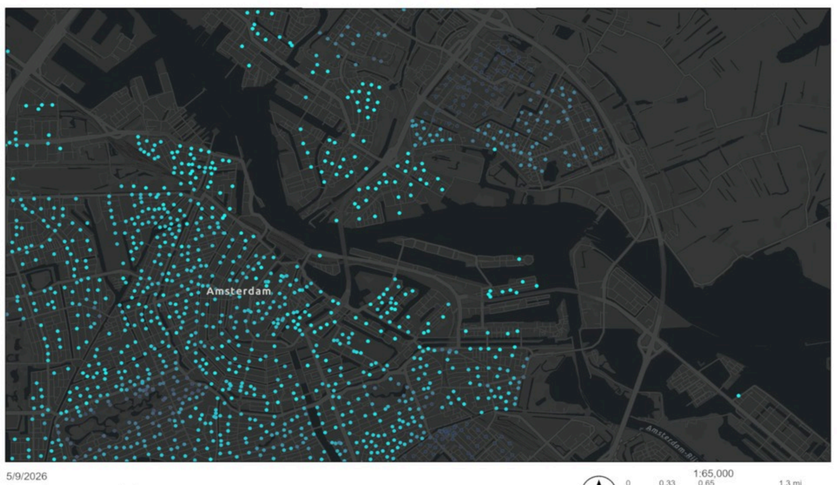
Amsterdam was selected as a global reference case for wateredge urbanism because of its advanced water-management system, dense river–harbour landscape and long history of adapting urban life to water. Its IJ Waterfront provides a compact 15 km corridor where flood resilience, blue-green infrastructure, public mobility and social equity can be studied together, offering a strong contrast to Qingdao's linear mountain–sea coastline.



Reason 01
Water Management



Reason 02
Climate Adaptation



Reason 03
Urban Regeneration



Comparative Urban Systems

ABEE4086 Urban Systems and Techniques

Xinyue Liu | Independent work

Dr. Alexander Zipprich [Module Convenor]

Amasterdam

Phase 1 | Systemic Mapping of the IJ-Waterfront Corridor

Amsterdam's IJ waterfront is not a single linear edge, but a layered river–harbour corridor shaped by flood resilience, blue-green infrastructure, and public access systems. Phase 1 maps these three systems together to reveal how water management, ecological networks, mobility, public facilities, and residential communities interact across the 15 km waterfront, forming the spatial foundation for later diagnosis and intervention.

System 01: Waterfront Protection



Amsterdam's waterfront resilience is formed by a distributed water-management network of dikes, polders, drainage facilities, groundwater-risk zones, and floodable landscapes. This layered system supports adaptation to flood and stormwater risks, but exposure and emergency access vary across waterfront districts.

System 02: Ecological Connection



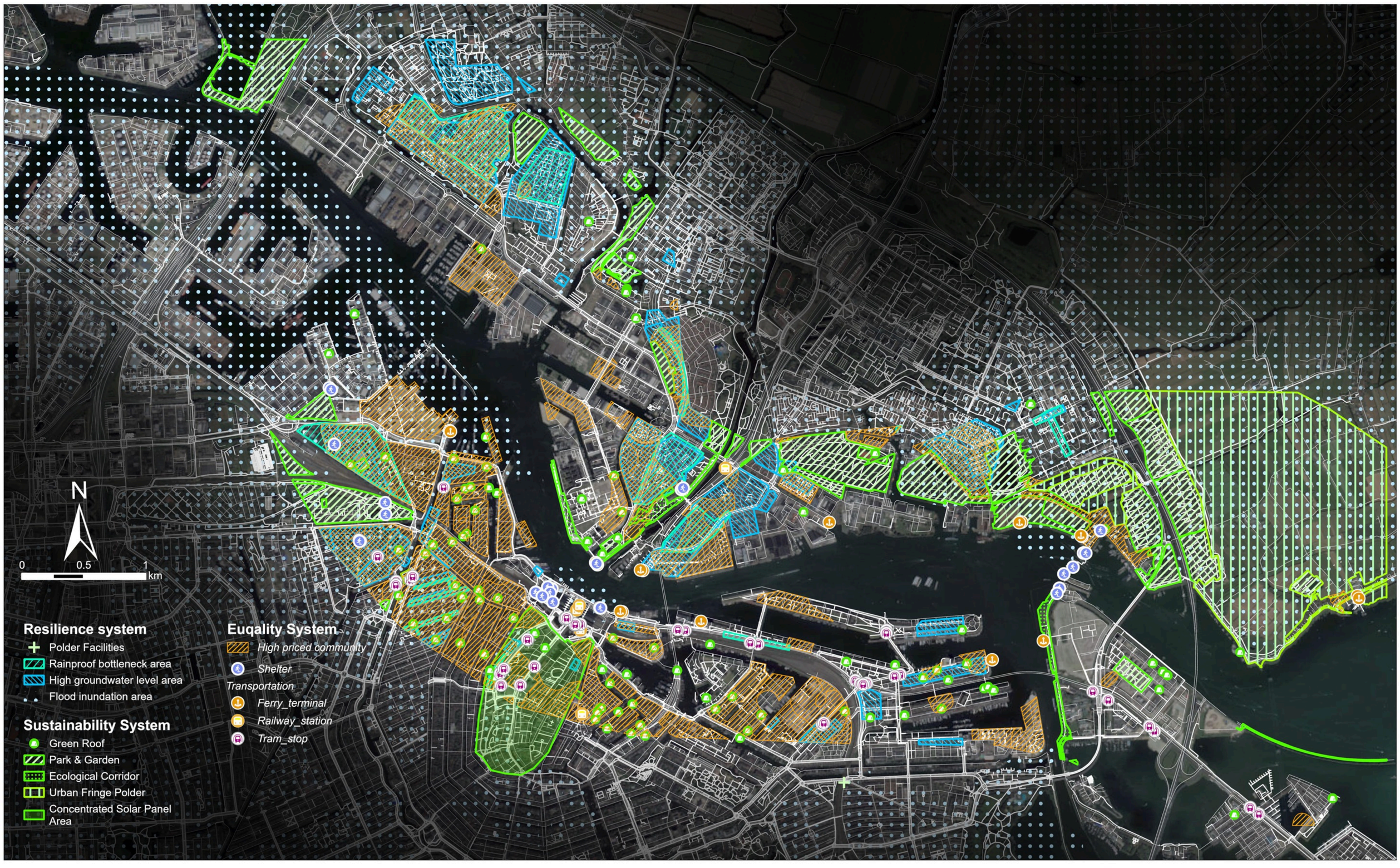
The sustainability system connects green roofs, parks, ecological corridors, recycling resources, and urban fringe polders into a blue-green network. These assets support biodiversity, cooling, rainwater retention, and recreation, yet their distribution remains fragmented in dense harbour and urban areas.

System 03: Shared Resources



The equity system reveals how public facilities, transport nodes, docks, cultural spaces, and high-value communities shape access to the waterfront. Central and higher-value areas are better served, while peripheral communities show weaker access to everyday public amenities.

System Mapping



Sources:

1.Fritz Pombom et al. (2025) 'Urban Design and Delta Landscape: A Case Study of Amsterdam', Architectural Journal/Jian ZhuXue Bao, (674), pp. 89-95. The doi: 10.19819 / j.carol carroll nki ISSN0529-1399.202503014.

2.Savini, F. (2025). Degrowth as ideology: Making values for the soil of Amsterdam. Environmental Values, 34(4-5), 327-347.

3.Picchi, P., Oudes, D., & Stremke, S. (2023). Regional Strategy, Municipality Plans and Site Designs for Energy Transition in Amsterdam, The Netherlands: How Sustainable Are Implementation Processes on Different Spatial Levels?. Sustainability, 15(7), 5876.

4.Sharma, S. E. (2023). Urban climate resilience under racial capitalism: governing pluvial flooding across Amsterdam and Dhaka. Geoforum, 145, 103817.

5.https://amsterdamsmartcity.com/

6.https://www.hollandsewaterlinies.nl/en

7.https://maps.amsterdam.nl/

Comparative Urban Systems

ABEE4086 Urban Systems and Techniques

Xinyue Liu | Independent work

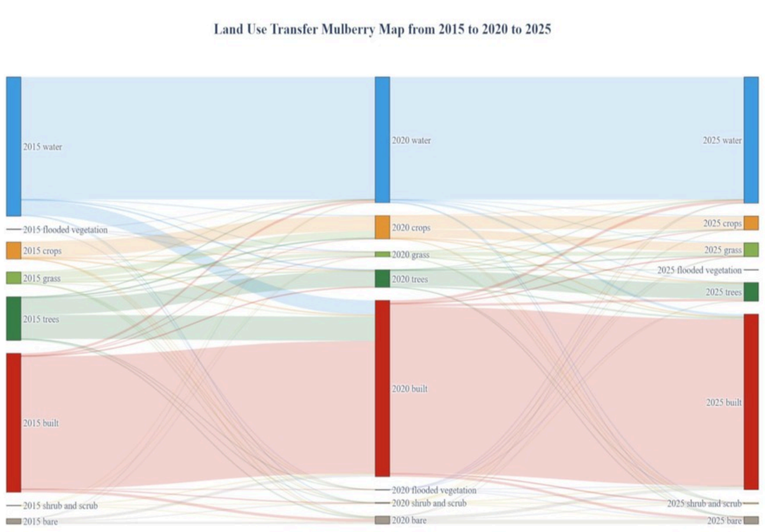
Dr. Alexander Zipprich [Module Convenor]

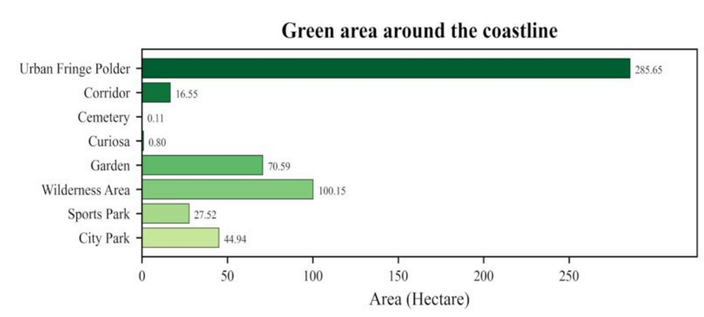
Amasterdam

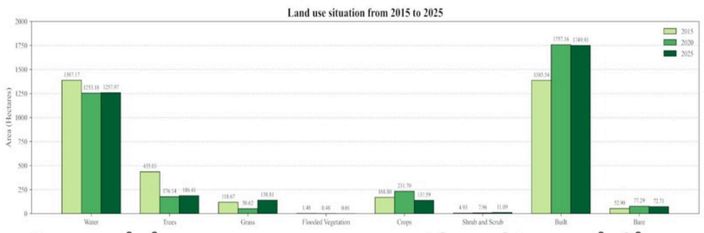
Phase 2 | Integrated Diagnosis

Amsterdam's waterfront corridor shows a mismatch between ecological infrastructure, everyday public facilities, and emergency shelter accessibility. Areas with better market value tend to receive stronger facility and shelter support, while lower-price and peripheral communities face combined disadvantages in green access, public services and safety infrastructure.

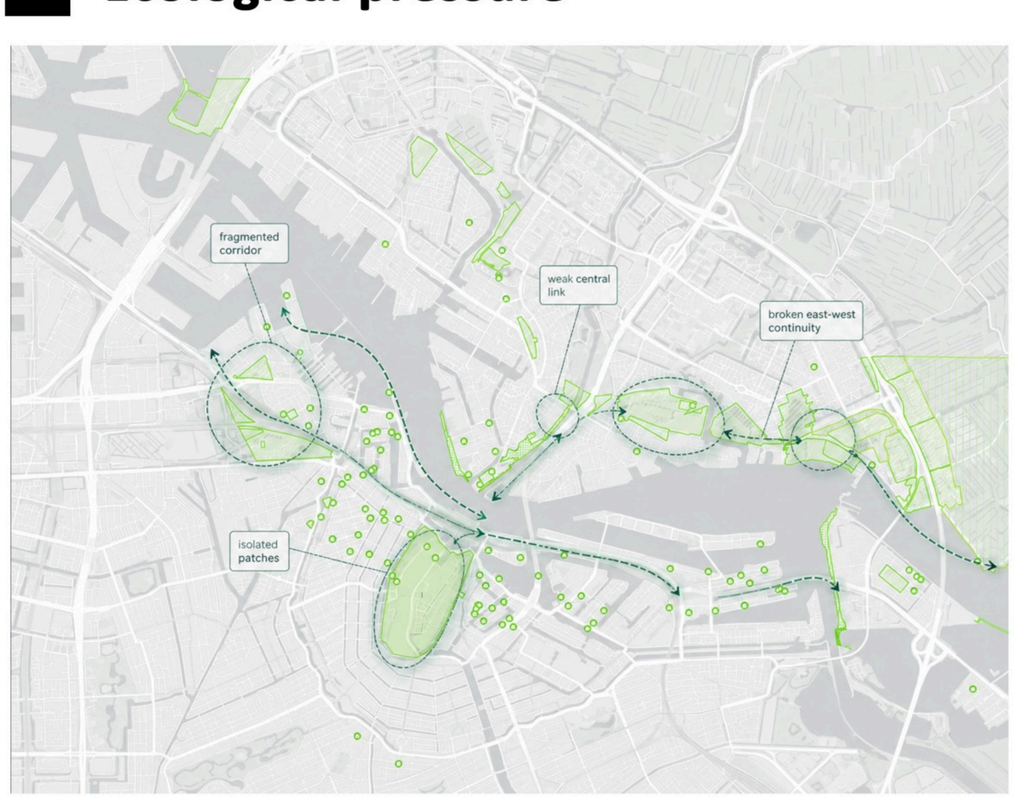
Ecological Transformation





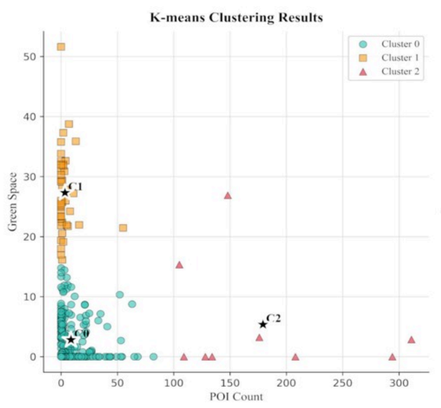


Ecological pressure

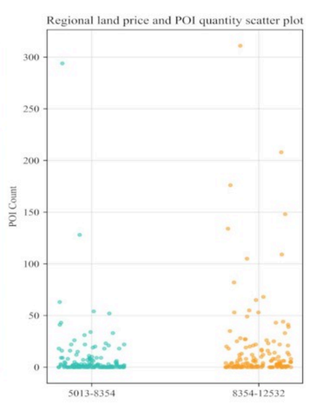


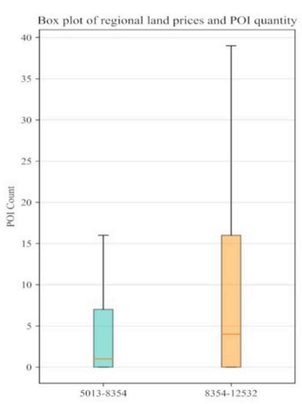
Q1: How can the waterfront corridor restore blue-green continuity while accommodating urban growth?

Social Spatial Inequality

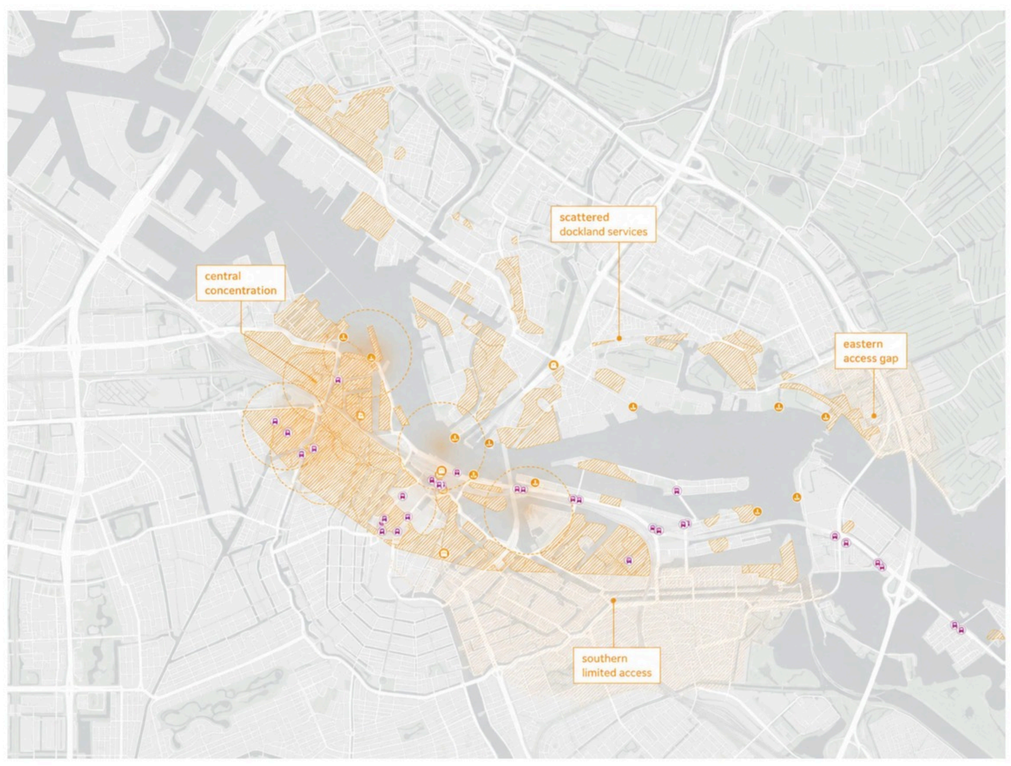






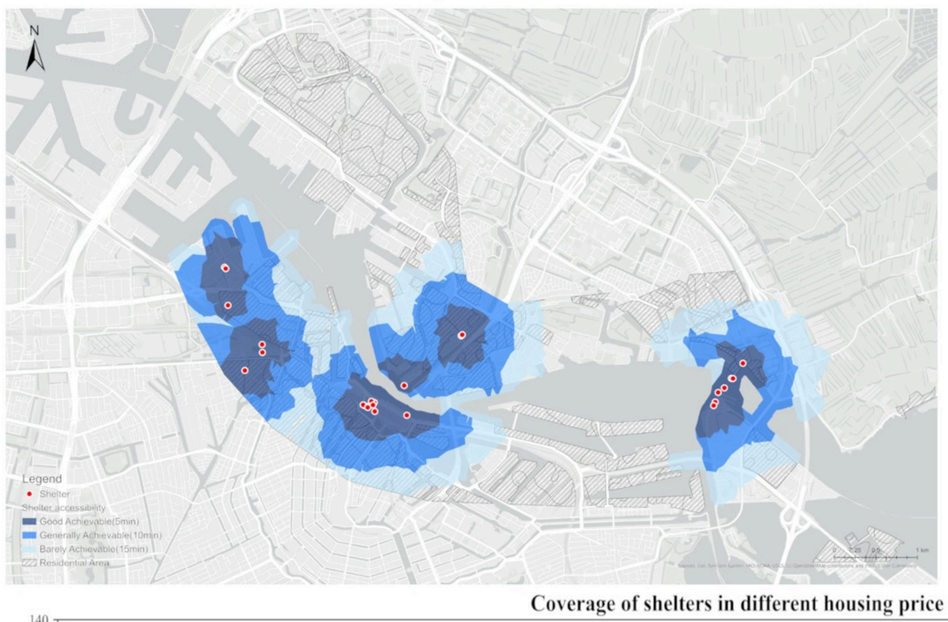


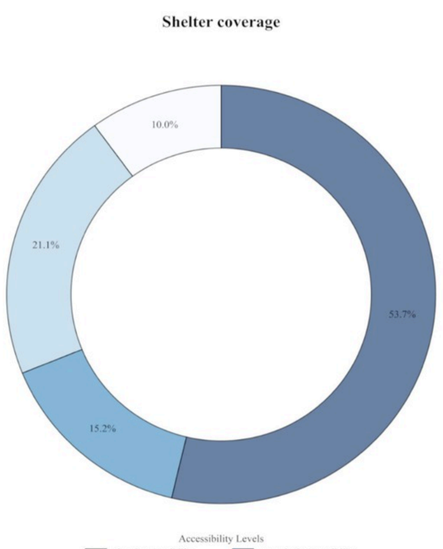
Facility inequality

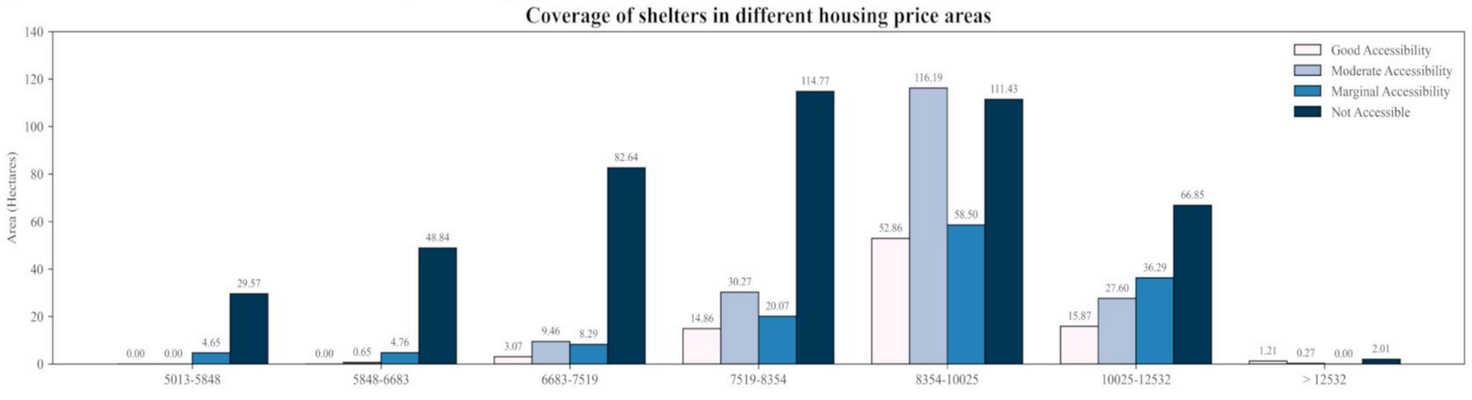


Q2: How can everyday public facilities and green spaces be redistributed toward underserved communities?

Shelter Accessibility and Equality

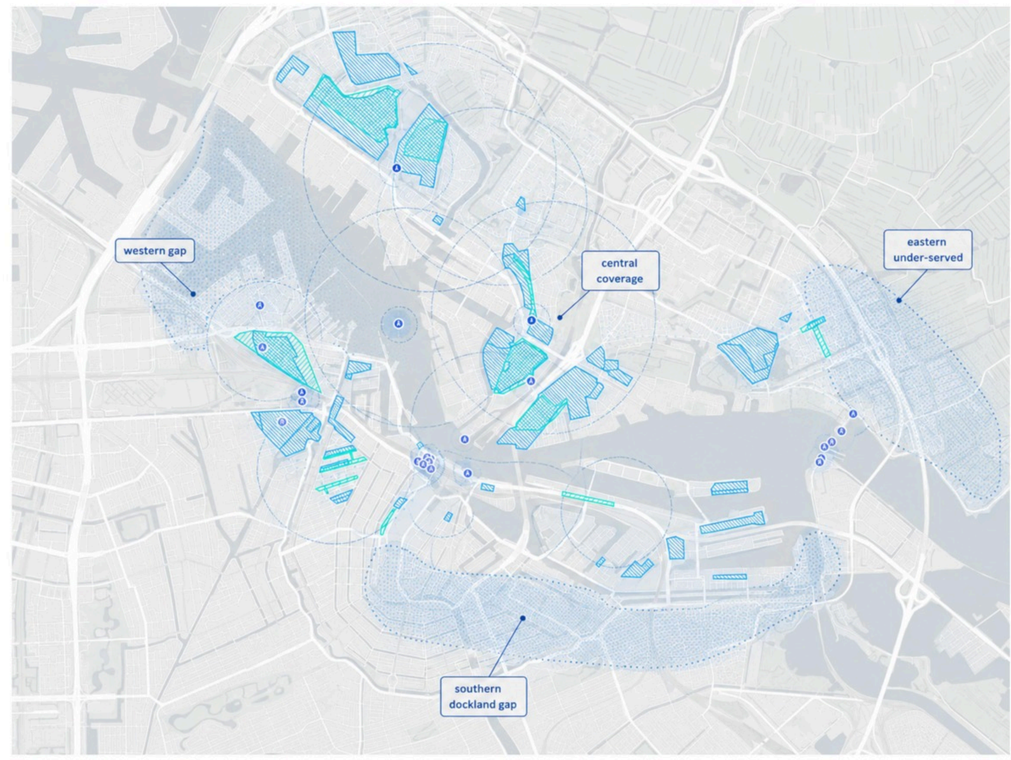






Q3: How can shelters become part of an everyday public network rather than emergency-only infrastructure?

Shelter gap



Sources:

1.Fritz Pombom et al. (2025) 'Urban Design and Delta Landscape: A Case Study of Amsterdam', Architectural Journal/Jian ZhuXue Bao, (674), pp. 89-95. The doi: 10.19819 / j.carol carroll nki ISSN0529-1399.202503014.

2.Savini, F. (2025). Degrowth as ideology: Making values for the soil of Amsterdam. Environmental Values, 34(4-5), 327-347.

3.Picchi, P., Oudes, D., & Stremke, S. (2023). Regional Strategy, Municipality Plans and Site Designs for Energy Transition in Amsterdam, The Netherlands: How Sustainable Are Implementation Processes on Different Spatial Levels?. Sustainability, 15(7), 5876.

4.Sharma, S. E. (2023). Urban climate resilience under racial capitalism: governing pluvial flooding across Amsterdam and Dhaka. Geoforum, 145, 103817.

5.https://amsterdamsmartcity.com/

6.https://www.hollandsewaterlinies.nl/en

7.https://maps.amsterdam.nl/

Comparative Urban Systems

ABEE4086 Urban Systems and Techniques

Xinyue Liu | Independent work

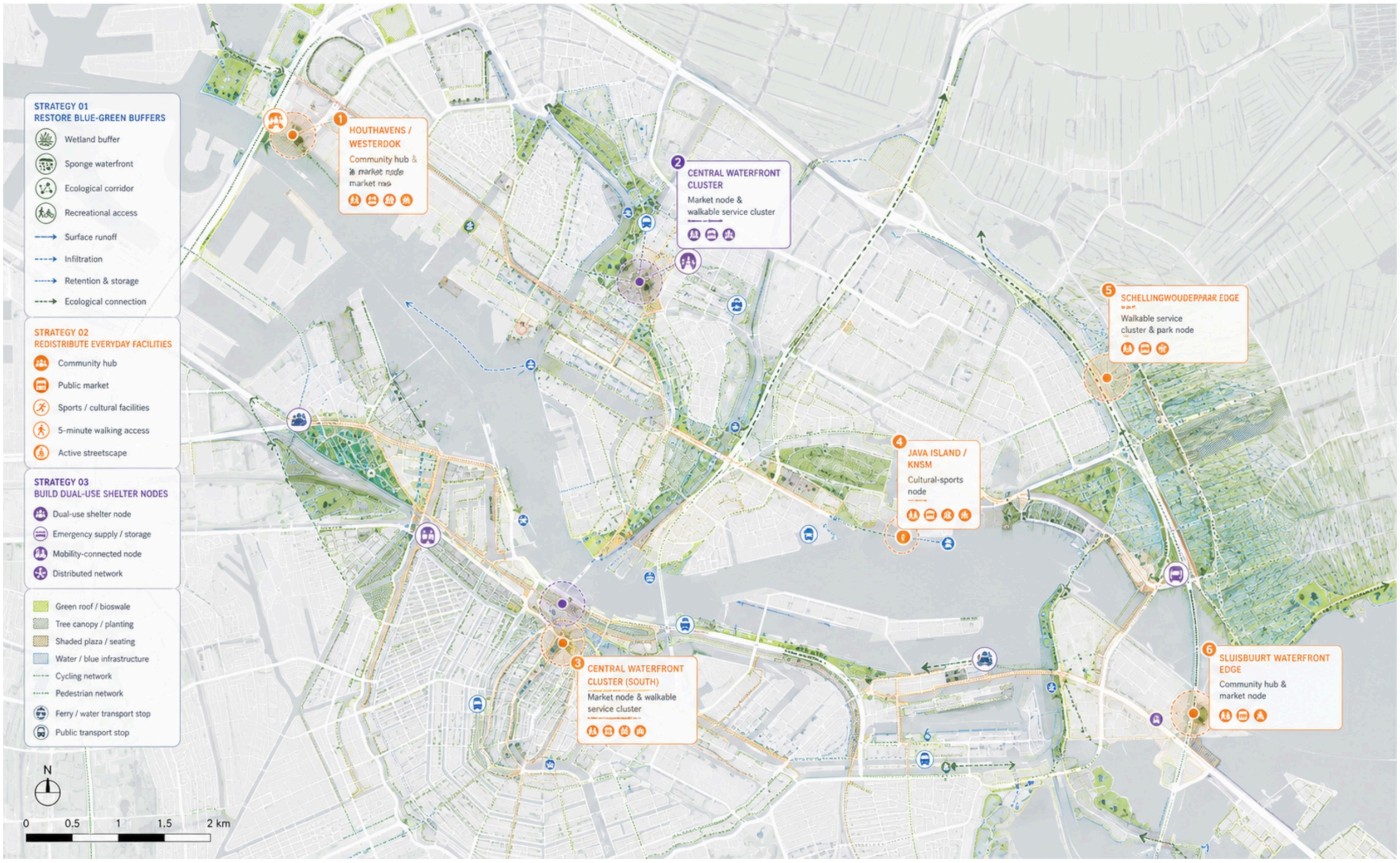
Dr. Alexander Zipprich [Module Convenor]

Amasterdam

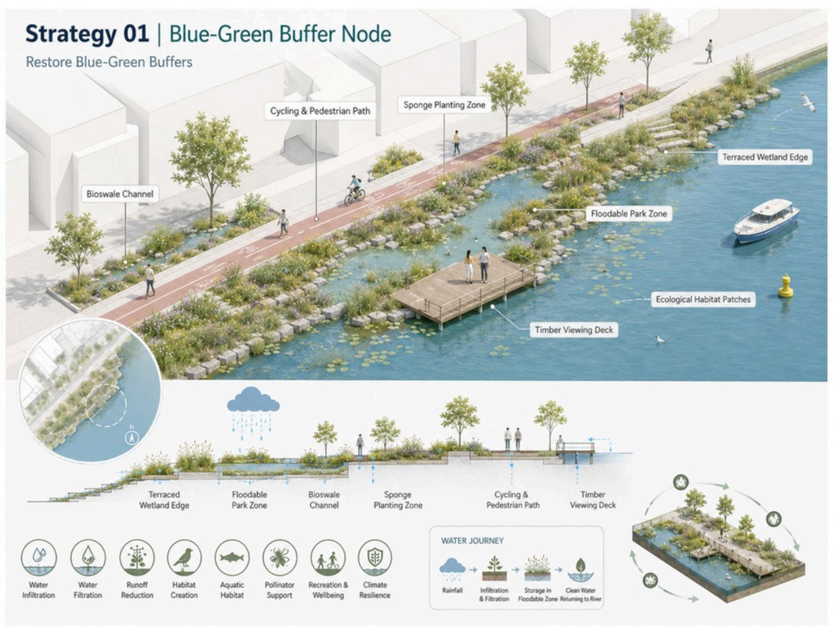
Phase 3| Equitable Blue-Green Shelter Network

Phase 3 proposes an equitable blue-green shelter network, combining ecological buffer restoration, everyday facility redistribution and distributed dual-use shelter nodes. By integrating shelters, green spaces, community public amenities, and waterfront areas, "emergency-resilient facilities" can serve as infrastructure for public life during everyday times as well.

Strategic Vision Board

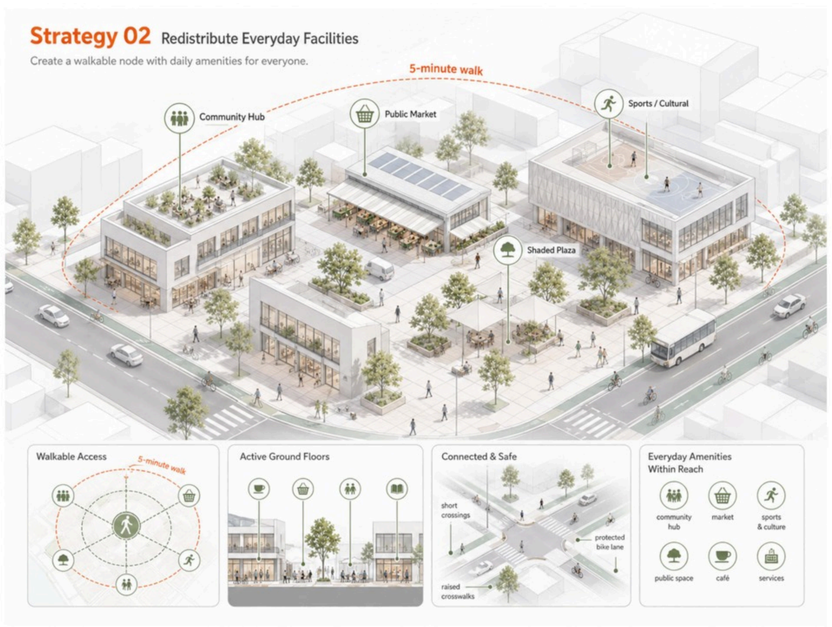


Strategy 01
Restore Blue-Green Buffers



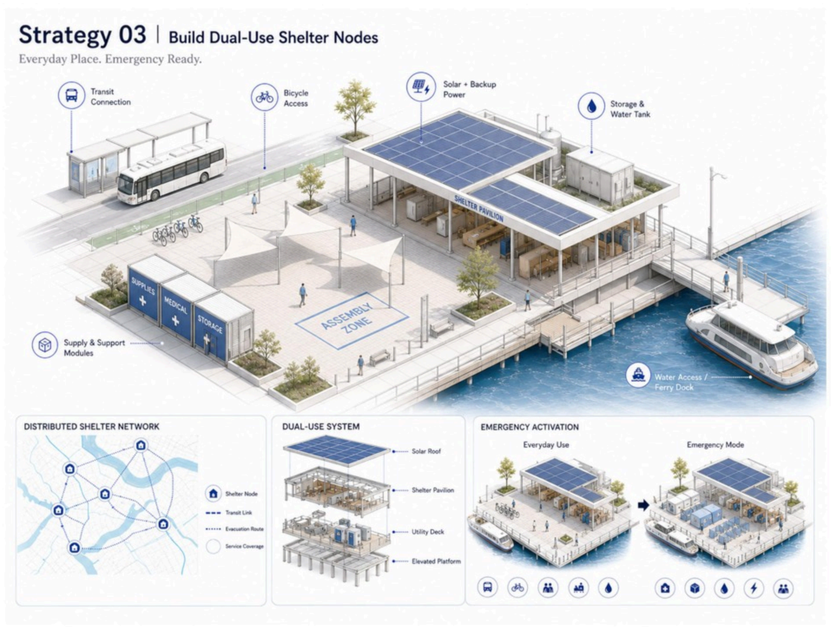
Restore wetland buffers, sponge waterfront edges, and ecological corridors to strengthen the waterfront's capacity to adapt to stormwater and high-water risks, while improving public access to water and recreation.

Strategy 02
Redistribute Everyday Facilities



Introduce community hubs, public markets, sports facilities, and cultural spaces in underserved waterfront neighbourhoods to create a 5-minute walking service network and improve spatial equity.

Strategy 03
Build Dual-Use Shelter Nodes



Integrate shelter nodes with community public spaces, mobility connections, and waterfront facilities, allowing them to function as open spaces and quickly transform into emergency shelters during extreme weather events.

Sources:
1.Fritz Pombom et al. (2025) 'Urban Design and Delta Landscape: A Case Study of Amsterdam', Architectural Journal/Jian ZhuXue Bao, (674), pp. 89-95. The doi: 10.19819 / j.carol carroll nki ISSN0529-1399.202503014.
2.Savini, F. (2025). Degrowth as ideology: Making values for the soil of Amsterdam. Environmental Values, 34(4-5), 327-347.
3.Picchi, P., Oudes, D., & Stremke, S. (2023). Regional Strategy, Municipality Plans and Site Designs for Energy Transition in Amsterdam, The Netherlands: How Sustainable Are Implementation Processes on Different Spatial Levels?. Sustainability, 15(7), 5876.
4.Sharma, S. E. (2023). Urban climate resilience under racial capitalism: governing pluvial flooding across Amsterdam and Dhaka. Geoforum, 145, 103817.
5.https://amsterdamsmartcity.com/
6.https://www.hollandsewaterlinies.nl/en
7.https://maps.amsterdam.nl/